# 1 Social Optimum: $3 \times 3$ Grid Network

## 1.1 Network

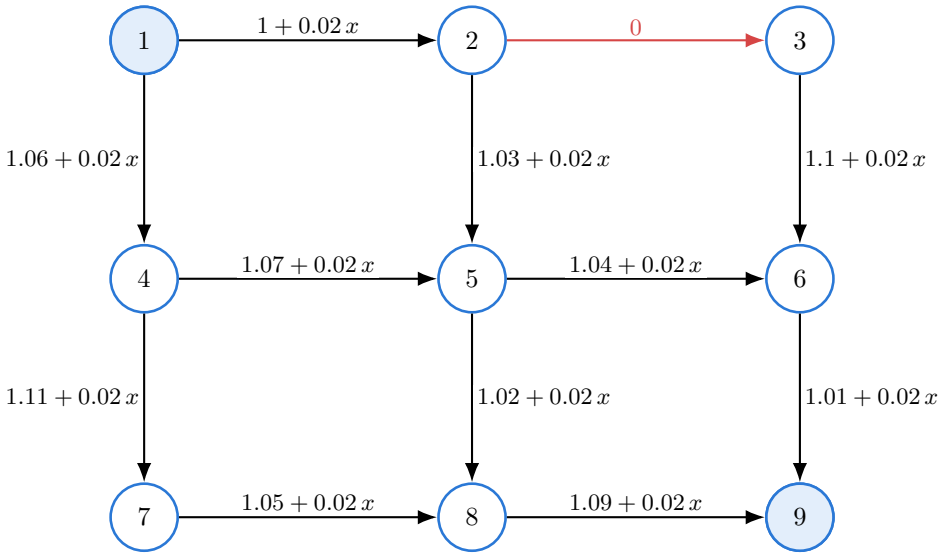



**Figure 1:** Grid network. $x$ is the number of agents on an edge in a timestep; $2 \to 3$ is free.

All agents travel $1 \to 9$, moving right or down, on one of six paths:

| Path | Nodes | Base cost $\sum a_e$ |
|---|---|---|
| RRDD | $1 \to 2 \to 3 \to 6 \to 9$ | 3.11 |
| RDRD | $1 \to 2 \to 5 \to 6 \to 9$ | 4.08 |
| RDDR | $1 \to 2 \to 5 \to 8 \to 9$ | 4.14 |
| DRRD | $1 \to 4 \to 5 \to 6 \to 9$ | 4.18 |
| DRDR | $1 \to 4 \to 5 \to 8 \to 9$ | 4.24 |
| DDRR | $1 \to 4 \to 7 \to 8 \to 9$ | 4.31 |

## 1.2 Derivation

The node in row $r$, column $c$ is reached at step $r + c$ on every path, so timing plays no role. With $x_e$ agents on edge $e$,

$$SC = \sum_e x_e\,(a_e + 0.02\,x_e).$$

At the optimum every used path has the same marginal cost $\sum_{e \in P}(a_e + 0.04\,x_e)$, and no unused path is cheaper. There is no simple closed form, so we solve it numerically: add agents one at a time along the path with the cheapest marginal cost, rerouting earlier agents when that helps (successive shortest paths). This is exact for convex costs.

**Data.** Edge costs and $N$ from `scenarios/grid_N20`, `scenarios/grid_N50` and `scenarios/grid`.

### 1.3 Results

| Path | $N = 20$ | $N = 50$ | $N = 80$ |
|---|---|---|---|
| RRDD | 15 | 28 | 39 |
| RDRD | 0 | 2 | 7 |
| RDDR | 0 | 0 | 0 |
| DRRD | 0 | 0 | 0 |
| DRDR | 3 | 10 | 17 |
| DDRR | 2 | 10 | 17 |
| $SC_{\text{opt}}$ | 83.01 | 256.58 | 481.58 |
| $SC_{\text{opt}}/N$ | 4.1505 | 5.1316 | 6.01975 |

**Table 1:** An optimal routing for each $N$. Edge loads are unique; the split into paths need not be.

- RRDD, the only path through the free edge, carries 75% of agents at $N = 20$, 56% at $N = 50$ and 49% at $N = 80$. As its other edges fill up, the optimum spreads agents across the grid.
- $N = 20$: the middle edges $2 \to 5$ and $5 \to 6$ are not worth using.
- $SC_{\text{opt}}/N$ rises from 4.15 to 6.02 as $N$ grows from 20 to 80.